

Wang Zhiyuan

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Education

Zhejiang University, Computer Science and Technology, Master's

2024.09-Present

Corporate Scholarship Outstanding Graduate Student Two first-author top-conference papers (ECCV 2026 and ACM MM 2026)

Wuhan University of Technology, Computer Science and Technology, Bachelor's

2020.10-2024.06

University Second-Class Scholarship College Merit Student 1 software copyright 1 patent

Professional Skills

Languages & Frameworks: Python, CUDA operator development, Triton operator development, operator fusion

LLM Inference: Familiar with LLM/Diffusion Model architectures and inference workflows

PyTorch Framework: Multi-Stream parallelism, CUDA Graph and Eager, torch.profiler end-to-end performance analysis

vllm/sglang: Distributed parallelism DP, TP, EP; MoE, KVcache, quantization, CUDA Graph, Prefix Cache and Chunked Prefill, speculative decoding.

Internship Experience

2026.07-Present Huawei - AI Infra

Responsible for speculative decoding deployment and performance optimization of DeepSeek-V4-Flash-DSPark on the Ascend 910/950 NPU platform: For the critical path of the Proposal stage, carried out multi-Stream parallel optimization and enabled the fused Attention operator. Designed and implemented splitting the Metadata preparation, Q and KV projections across different Streams, with Event synchronization guaranteeing data dependencies before the fused Attention; among them, the Metadata Stream overlaps with the Embedding and TargetHidden projections, and the MLA Stream executes the Target Context/Draft Block KV projection in parallel, reducing Proposal latency. Compared with MTP1, equivalent latency decreased by 54.3%, corresponding to about 2.19x equivalent generation efficiency.

Migrated the overlap scheme of sglang's scheduler: During speculative decoding, assumes full acceptance in the verify stage, triggers the asynchronous device-to-host result copy-back in advance, and achieves parallelism between D2H and draft-model decoding computation, as well as between next-stage scheduling and forward computation. This eliminated the hardware idle bubbles caused in traditional speculative decoding by having to wait for verification to finish before starting data copy-back, and eliminated the serial dependency of CPU scheduling having to wait for GPU execution to complete. End-to-end TPOT performance improved by 5%.

Project Experience

Diffusion Transformer Inference Acceleration & Cache Optimization System R&D: Against the background of high inference latency caused by the iterative denoising of Diffusion Transformers, existing Cache-then-Forecast methods rely on feature trajectory prediction and suffer from error accumulation and degraded generation quality at high acceleration ratios. Studied a training-free acceleration method based on Feature Cache, specifically proposing the DriftCache diffusion-model acceleration framework, which replaces layer-level feature caching with a Global Drift aligned by predicting the ODE, avoiding the accumulation of deep-layer Transformer errors; designed a parameter-free Barycentric Lagrange extrapolator and a Phase Mapping mechanism to improve the stability of cache skip-step prediction and achieve high-ratio inference acceleration. Validated on A100 GPUs with FLUX.1-dev and HunyuanVideo, achieving up to about 5x inference acceleration; at 4.97x acceleration, FLUX.1-dev improves ImageReward by 64%, reduces LPIPS by 52%, and reduces FID on the ImageNet generation task by 58%. **The related algorithms have been integrated into the open-source inference repository cann-recipes-infer, achieving deployment optimization of FLUX.1-dev and HunyuanVideo on the Ascend NPU platform. The related results were published as first author at the top conference ECCV 2026.**

Multimodal Diffusion Model Soft-Hardware Co-Acceleration System Development: Multimodal diffusion models (DMLLMs) with fixed-length generation lead to substantial repeated computation on [EOS] Padding tokens, causing serious inference overhead; meanwhile, dynamic sequence trimming breaks the static execution efficiency of CUDA Graph, limiting actual deployment performance. Proposed a soft-hardware co-acceleration framework which, by analyzing the MLP activation sparsity of early layers at Step-0, automatically locates the effective generation length and prunes redundant Padding tokens; designed Padding Waste-aware Hybrid Execution, dynamically selecting among Static Graph, Varlen Graph and Eager execution paths according to sequence length, balancing computation reduction with high-throughput inference. Validated on A100 GPUs with mainstream Diffusion MLLMs such as LaViDa-LLaDA and MMaDA, with throughput improving by up to about 31x; model performance is maintained across 9 multimodal tasks such as MME, MMMU, DocVQA and ChartQA, and cross-modal noise interference is reduced by cutting redundant suffix tokens. **The related results were published as first author at the top conference ACM MM 2026.**

High-Performance GPU Operator Development & Optimization: Developed and optimized CUDA operators such as LayerNorm, RMSNorm, FC, Softmax and Reduce, supporting multiple Shapes and dtypes, and connected the PyTorch interface to achieve Native/PyTorch accuracy alignment; based on the Roofline model and Nsight Compute to analyze bottlenecks, combined with techniques such as Tiling, vectorized memory access, Shared Memory optimization and Warp-level Reduce, raising operator compute/bandwidth utilization to 70%+.

Profiling-Based End-to-End Model Performance Optimization: Based on torch.profiler, analyzed the model execution Trace to locate bottlenecks in Kernel Launch, Stream idle bubbles and communication-computation overlap; by eliminating synchronization points and introducing Pinned Memory and a Non-blocking asynchronous transfer mechanism, the Trace shows the synchronization points and the corresponding stream idle bubbles eliminated, achieving a 1.5% end-to-end inference speedup; integrated fused operators such as APEX LayerNorm, RMSNorm, RoPE and AdamW, improving end-to-end model performance by 13%.

Int8 KV Cache Quantization Optimization Validation: For Qwen models, conducted a KV Cache quantization feasibility analysis, systematically evaluating the per-token/per-channel/per-head distribution characteristics and determining a per-channel quantization granularity; implemented the pre-Attention KV Cache quantization/dequantization flow and Page Attention INT8 GEMM simulation validation, achieving near-lossless accuracy on tasks such as Human and GPQA, providing a validation basis for the performance optimization of static KV Cache quantization.